Safexpress Centre for Data, Learning and Decision Sciences

# Experimental AI Subseasonal to Seasonal Forecast Guidance for India

Forecast initialized 9 September 2026

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| **Forecast period** | 9 September-6 October 2026 |
| **Forecast range** | Weeks 1-4 |
| **Reference period** | ERA5 reanalysis and model climatology, 2002-2021 |
| **Forecast product** | IFS operational proxy, 100-member ensemble mean |
| **Cyclogenesis outlook** | KGPP environmental favorability index, Days 1-15 |
| **Status** | Experimental research guidance |

## Executive Summary

The 9 September 2026 guidance does not show a simple, steady transition toward monsoon withdrawal. Instead, it depicts three distinct phases: a west-focused wet anomaly during Week 1, widespread drying with a localized southwest-peninsula wet signal during Week 2, and a renewed wet anomaly across much of peninsular, central, and eastern India during Week 3. By Week 4, the continental signal weakens and becomes less organized. During Week 1, rainfall is above normal over Gujarat, adjoining Rajasthan, western Madhya Pradesh, and parts of Maharashtra, while negative anomalies cover eastern India, Northeast India, and much of the Bay of Bengal. Week 2 is drier than normal across most of the country, with a notable positive anomaly over the southwest peninsula. Week 3 reverses this pattern: above-normal rainfall expands across much of the peninsula and central India and extends into eastern India and the Himalayan foothills. Week 4 is mostly near normal, with a weak dry corridor from central India toward the northern Bay of Bengal and small wet anomalies in the far north and Northeast.

Temperature and circulation fields support this changing pattern. Broad cool anomalies cover northern and central India during Week 1. Central India becomes warmer during Week 2 as rainfall declines, whereas the renewed Week 3 rainfall coincides with cooler conditions over much of the southern peninsula. Low-level winds are weaker than normal over much of India and the adjoining seas in Weeks 1-2. In Week 3, stronger-than-normal westerly to southwesterly flow develops across the southern peninsula and nearby ocean, consistent with renewed moisture transport. Confidence is highest in Week 1, moderate in Week 2, and lower in Weeks 3-4.

**Cyclogenesis signal** The KGPP maps identify changing areas where the large-scale environment may support tropical cyclone formation. They do not give the probability of cyclone formation, a predicted track, or an intensity forecast.

## 1 Methodology

The weekly guidance uses the IFS operational initialization from 9 September 2026 and a 100-member ensemble mean. Weekly rainfall, 2 m temperature, and 850 hPa wind fields are shown for four seven-day periods. Anomalies are departures from the model climatology for the corresponding season and forecast lead during 2002-2021. The displayed maps use visual interpolation from 1.5 degrees to 0.25 degrees; values remain on the native model grid. The cyclogenesis outlook uses KGPP, an environmental favorability index that combines atmospheric ingredients associated with tropical cyclone formation, including moist and unstable air, low-level rotation, and sufficiently weak vertical wind shear. Higher KGPP values indicate a more supportive environment. They are not percentages and must not be interpreted as the probability that a cyclone will form.

## 2 Rainfall Forecast

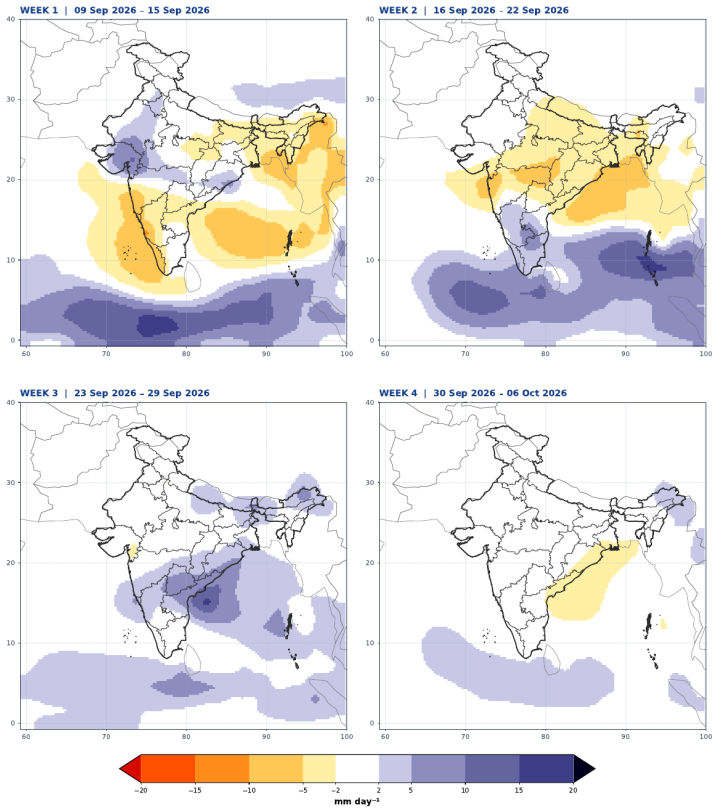



Figure 1. Weekly mean rainfall anomaly over India for Weeks 1-4, IFS operational proxy initialized 9 September 2026. Panels cover 9-15 September, 16-22 September, 23-29 September, and 30 September-6 October 2026. Anomalies are departures from the model climatology for the corresponding season and forecast lead during 2002-2021.

**Week 1  9 September to 15 September 2026:** Week 1 shows the strongest positive continental rainfall anomaly over western and west-central India. Above-normal rainfall is centered on Gujarat and adjoining parts of Rajasthan, western Madhya Pradesh, and Maharashtra. The

absolute rainfall field maintains a broad rain belt across central and eastern India, but the anomaly field indicates that rainfall over eastern India, Northeast India, and much of the Bay of Bengal is below the model climatology. Southern peninsular India is mostly near normal to drier than normal.

**Week 2  16 September to 22 September 2026:** Negative rainfall anomalies expand across most of northern, central, eastern, and peninsular India during Week 2. The clearest exception is the southwest peninsula, where a positive anomaly is centered over parts of Karnataka and Kerala and the adjoining eastern Arabian Sea. Positive anomalies also appear across the southern Indian Ocean, while the northern Bay of Bengal remains drier than normal. This week represents the most spatially extensive dry signal over India in the four-week forecast.

**Week 3  23 September to 29 September 2026:** Week 3 shows a marked reversal from Week 2. Above-normal rainfall spreads across much of peninsular and central India, including Maharashtra, Karnataka, Telangana, Andhra Pradesh, southern and eastern Madhya Pradesh, Chhattisgarh, Odisha, and adjoining areas. Positive anomalies also extend along parts of the Himalayan foothills and into Northeast India. The breadth of this wet anomaly argues against interpreting the Week 2 drying alone as the onset of a sustained withdrawal phase.

**Week 4  30 September to 6 October 2026:** By Week 4, rainfall anomalies weaken considerably. Much of India is close to the model climatology. A weak negative anomaly extends from parts of central and east-central India toward the northern Bay of Bengal, while small positive anomalies remain over the far north, portions of Northeast India, and the southern Indian Ocean. The weak ensemble-mean pattern at this lead should be interpreted cautiously because averaging can smooth individual synoptic events.

**3 Wind and Circulation Forecast**

The 850 hPa fields provide the dynamical context for the rainfall evolution. The anomaly shading shows the difference in weekly wind speed from climatology, while the arrows show anomalous zonal and meridional wind components. Brown shading therefore means weaker wind speed than normal; it does not by itself identify wind direction.

**Week 1:** Low-level wind speeds are weaker than normal across much of peninsular and central India and the adjoining Indian Ocean. Easterly anomaly vectors appear across large parts of the peninsula and southern ocean, indicating reduced climatological westerly flow. Over western and central India, the vector pattern includes convergence and curvature near the Week 1 wet-anomaly region, which is consistent with locally enhanced rainfall despite the broader weakening of low-level winds.

**Week 2:** The negative wind-speed anomaly strengthens across the peninsula and both adjoining seas. Anomalous easterly flow remains prominent across southern and central latitudes. This configuration is consistent with the widespread negative rainfall anomaly over most of India. The localized wet anomaly over the southwest peninsula occurs within a more regional circulation feature and should not be read as a nationwide monsoon revival.

**Week 3:** A substantial circulation change occurs during Week 3. Wind speeds become stronger than normal across the southern peninsula and nearby Indian Ocean, with anomalous westerly to southwesterly flow directed toward southern and central India. This provides a plausible moisture-transport pathway for the renewed positive rainfall anomaly across the peninsula, central India, and parts of eastern India.

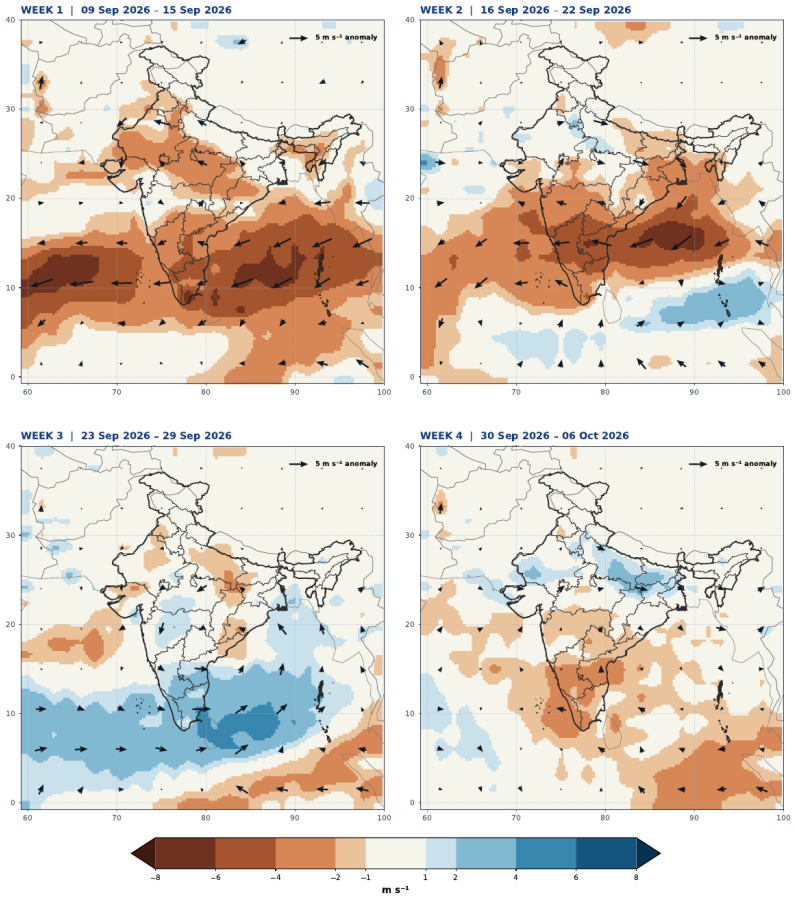



Figure 2. Weekly mean 850 hPa wind anomaly for Weeks 1-4, initialized 9 September 2026. Shading represents weekly wind-speed anomalies, and arrows represent anomalous U and V wind components relative to 2002-2021.

**Week 4:** The Week 3 strengthening does not persist clearly into Week 4. Wind anomalies weaken and become spatially mixed, with below-normal wind speeds again evident over parts of southern India and the surrounding ocean. The circulation and rainfall signals are both weak and fragmented, limiting confidence in any specific withdrawal or revival interpretation at this lead.

## 4 Temperature Forecast

**Week 1  9 September to 15 September 2026:** Below-normal temperatures dominate northern, central, and eastern India during Week 1, including much of the Indo-Gangetic Plain and the central Indian belt. Warmer-than-normal conditions are concentrated over the southern peninsula and appear locally along the western Himalayan region. The broad cool anomaly over central and northern India is consistent with cloud cover and rainfall across the monsoon belt, even where rainfall departures are not uniformly positive.

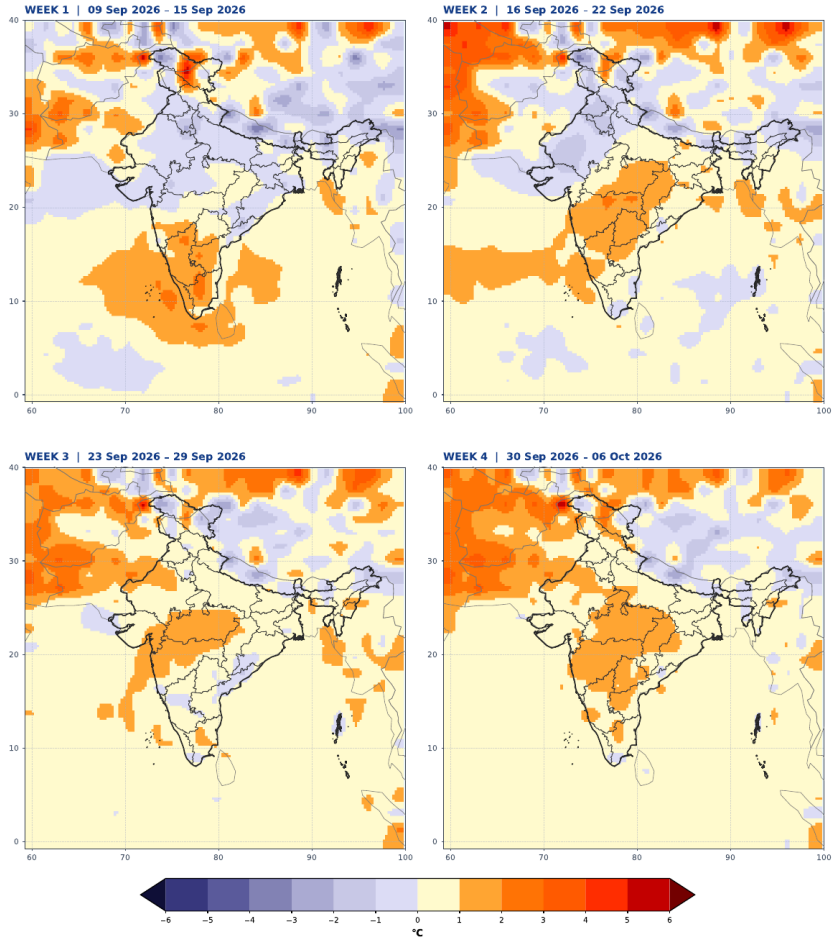



Figure 3. Weekly mean 2 m temperature anomaly over India for Weeks 1-4, IFS operational proxy initialized 9 September 2026. Anomalies are relative to the model climatology for the corresponding season and forecast lead during 2002-2021.

**Week 2  16 September to 22 September 2026:** Positive temperature anomalies develop across much of central India and parts of the eastern peninsula as rainfall becomes broadly deficient. Cooler-than-normal conditions persist over northwestern and northern India, parts of the Himalayan foothills, Northeast India, and the far southern peninsula. The warm central-India signal is dynamically consistent with reduced rainfall and cloud cover.

**Week 3  23 September to 29 September 2026:** Warm anomalies persist over western and central India and appear locally over eastern and northeastern India. In contrast, much of the southern peninsula turns cooler than normal as the rainfall anomaly becomes positive and low-level moisture transport strengthens. This north-south contrast is one of the clearest temperature signals in Week 3.

**Week 4  30 September to 6 October 2026:** Warmer-than-normal conditions expand across western, central, and much of peninsular India during Week 4. Cooler anomalies remain concentrated along the Himalayan belt and parts of Northeast India. Because rainfall and circulation anomalies are weak in Week 4, the broad warm signal should be treated as a seasonal-scale ensemble tendency rather than evidence of a specific weather event.

## 5 Cyclogenesis Favorability Outlook

The KGPP product shows a migrating pattern of environmental favorability during the 15-day outlook. On 11 September, elevated values extend across eastern India and the adjoining northern Bay of Bengal. By 13-17 September, the most concentrated signal shifts toward western India and the northeastern Arabian Sea, with additional patchy favorability near the Bay of Bengal and Southeast Asia. Around 20 September, a broader area of elevated KGPP appears over the central and southern Bay of Bengal. By 25 September, the strongest signal is again concentrated over the northern Bay of Bengal and adjoining eastern India and Bangladesh.

This sequence identifies periods and basins that warrant closer monitoring, but it does not predict that a cyclone will form or follow a particular track. A high index can occur without genesis, and a tropical disturbance can still develop outside the strongest index area. Confidence also decreases with forecast lead, especially for the Day 10 and Day 15 panels. Any operational assessment should use official bulletins and ensemble disturbance tracking in addition to this environmental index.

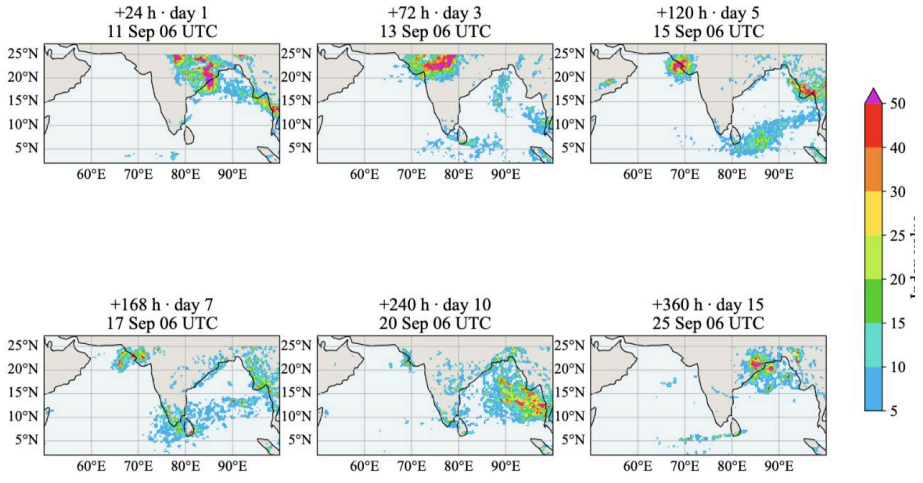


Figure 4. KGPP cyclogenesis environmental favorability index at Days 1, 3, 5, 7, 10, and 15, valid at 06 UTC on 11, 13, 15, 17, 20, and 25 September 2026. Higher values indicate a more supportive large-scale environment; they are not cyclone probabilities.

**Cyclogenesis Signals to Monitor**

- 11 September: eastern India and the adjoining northern Bay of Bengal.
- 13-17 September: western India and the northeastern Arabian Sea, with secondary patchy signals farther east.
- 20 September: central and southern Bay of Bengal.
- 25 September: northern Bay of Bengal, eastern India, and Bangladesh.

## 6 Key S2S Signals

- **Western wet anomaly in Week 1.** Above-normal rainfall is centered over Gujarat and adjoining west-central India, while eastern India and much of the Bay of Bengal are drier than normal.
- **Broad drying in Week 2.** Negative rainfall anomalies cover most of India, with a localized wet exception over the southwest peninsula.
- **Renewed rainfall in Week 3.** Positive anomalies expand across much of peninsular, central, and eastern India and reach the Himalayan foothills and Northeast India.
- **Weak Week 4 signal.** Rainfall and circulation anomalies become small and fragmented, and forecast confidence is correspondingly lower.
- **Circulation interruption and recovery.** Weaker-than-normal low-level winds dominate Weeks 1-2; stronger westerly to southwesterly anomalies return over southern India and nearby waters in Week 3.
- **Temperature response.** Central India warms during the dry Week 2, while the southern peninsula cools during the wetter Week 3.
- **Changing cyclogenesis environment.** KGPP favorability shifts from the northern Bay and eastern India to the northeastern Arabian Sea, then returns to the Bay later in the 15-day period.

## 7 Interpretation

The 9 September forecast suggests a variable late-monsoon regime rather than a smooth progression toward withdrawal. Week 1 favors western and west-central India, while eastern India and the Bay of Bengal are relatively dry. Week 2 brings the most widespread negative rainfall anomaly and the clearest weakening of low-level flow. If considered alone, that pattern could resemble an early withdrawal-like transition.

Week 3 materially changes the interpretation. Above-normal rainfall returns across a broad area from the southern peninsula through central and eastern India, accompanied by stronger westerly to southwesterly low-level flow over southern India and the adjacent ocean. This circulation-rainfall coupling is more consistent with a temporary revival or renewed moisture transport than with uninterrupted withdrawal. Week 4 then returns to a weak, poorly organized ensemble-mean signal.

The cyclogenesis index provides an additional short-range monitoring layer. It highlights a favorable environment near eastern India and the northern Bay early in the period, over western India and the northeastern Arabian Sea around Days 3-7, and over the Bay again around Days 10-15. These areas may help identify where closer monitoring is warranted, but the index cannot establish genesis, track, landfall, or intensity. Official national meteorological warnings remain authoritative.

Overall confidence is highest for Week 1, moderate for Week 2, and lower for Weeks 3-4. The Week 3 revival is a notable ensemble-mean signal, but its timing and spatial extent should be checked against subsequent initializations. No statement in this guidance should be treated as confirmation of formal southwest-monsoon withdrawal or of tropical cyclone formation.

**Data and Product Notes**

Forecast products: https://s2s.scdlds.com/

Rainfall, temperature, and 850 hPa wind panels: IFS operational proxy initialized 9 September 2026, 100-member ensemble mean. Reference period: 2002-2021. Cyclogenesis panel: supplied KGPP outlook valid 11-25 September 2026. This document is experimental research guidance and is not an operational warning.